Name: Gaurang Vaghela
Rollno: BEAD-22561
CI Practical 2
Problem Statement: Design a distributed application using RMI for remote computation where client submits two strings to the server and server returns the concatenation of the given strings.

Server.py

```
import Pyro4

@Pyro4.expose
class StringConcatenationServer:
    def concatenate_strings(self, str1, str2):
        result = str1 + str2
        return result

def main():
    daemon = Pyro4.Daemon()
    ns = Pyro4.locateNS()

    server = StringConcatenationServer()

    uri = daemon.register(server)
    ns.register("string.concatenation", uri)
    print("Server URI:", uri)
    with open("server_uri.txt", "w") as f:
        f.write(str(uri))
    daemon.requestLoop()

if __name__ == "__main__":
    main()
```

Client.py

```
import Pyro4

def main():
    with open("server_uri.txt", "r") as f:
        uri = f.read()
    server = Pyro4.Proxy(uri)  # Connect to the remote server
    str1 = input("Enter the first string: ")
    str2 = input("Enter the second string: ")
    result = server.concatenate_strings(str1, str2)
    print("Concatenated Result:", result)

if __name__ == "__main__":
    main()
```

OUTPUT:

Server:

```
C:\WINDOWS\System32\cmd.e   Windows PowerShell   C:\WINDOWS\System32\cmd.

D:\College\SEM 8\CL 3\CI\CI2>Pyro4-ns
Not starting broadcast server for localhost.
NS running on localhost:9090 (127.0.0.1)
Warning: HMAC key not set. Anyone can connect to this server!
URI = PYRO:Pyro.NameServer@localhost:9090
```

```
C:\WINDOWS\System32\cmd.e   Windows PowerShell   C:\WINDOWS\System32\cmd.e

PS D:\College\SEM 8\CL 3\CI\CI2> python Server.py
Server URI: PYRO:obj_daf63d1a1f404416b7a251a728e9aa92@localhost:54624
```

Client:

```
C:\WINDOWS\System32\cmd.   Windows PowerShell

Microsoft Windows [Version 10.0.26200.7623]
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D:\College\SEM 8\CL 3\CI\CI2>Python Client.py
Enter the first string: Hello
Enter the second string: World!
Concatenated Result: Hello World!

D:\College\SEM 8\CL 3\CI\CI2>
```